

GROUP TRISECTIONS AND SMOOTHLY KNOTTED SURFACES

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ABSTRACT. Trisector Workshop lecture notes.

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¹Based on lectures by Sarah Blackwell [2].

Part 1. Group Trisections and Smoothly Knotted Surfaces²

1. HANDLEBODY EPIMORPHISMS AND GROUP TRISECTIONS

Cut systems translate three-dimensional handlebodies into epimorphisms from surface groups to free groups, and pushout cubes of three such maps encode smooth four-manifold trisections. All fundamental groups are based, with basepoint paths suppressed, so changing a path conjugates the corresponding homomorphism. Throughout

$$S_g := \pi_1(\Sigma_g, *) = \left\langle x_1, y_1, \dots, x_g, y_g \mid \prod_{j=1}^g [x_j, y_j] = 1 \right\rangle,$$

$$F_r := \langle d_1, \dots, d_r \rangle, \quad [u, v] := uvu^{-1}v^{-1}, \quad F_0 := 1,$$

where Σ_g is a closed connected oriented surface.

1.1. Cut systems and splitting epimorphisms.

Definition 1.1.1 (Cut system and associated handlebody). A cut system on Σ_g is an ordered tuple of pairwise disjoint essential simple closed curves satisfying

$$\delta = (\delta_1, \dots, \delta_g), \quad \Sigma_g \setminus \bigcup_{j=1}^g \delta_j \text{ connected.}$$

Attaching $D^2 \times [-1, 1]$ along annular neighborhoods of $\delta_j \times \{1\} \subset \Sigma_g \times \{1\}$ and capping the spherical boundary gives

$$H_\delta := \left(\Sigma_g \times [0, 1] \cup \bigcup_{j=1}^g D^2 \times [-1, 1] \right) \cup_{S^2} B^3, \quad \partial H_\delta = \Sigma_g \times \{0\}.$$

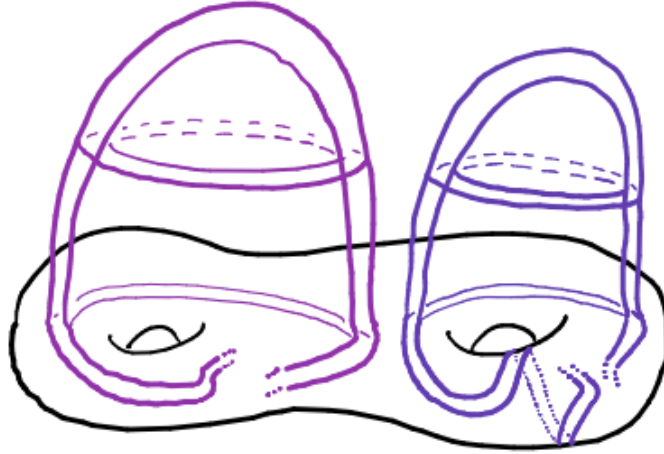


FIGURE 1. A cut system on Σ_2 bounds a complete disk system.

²Based on lectures by Sarah Blackwell [2].

Proposition 1.1.2 (Handlebody quotient). *Every cut system determines*

$$H_\delta \cong \natural^g(S^1 \times D^2), \quad (i_\delta)_* : S_g \twoheadrightarrow \pi_1(H_\delta) \cong F_g, \quad \ker(i_\delta)_* = \langle\langle [\delta_1], \dots, [\delta_g] \rangle\rangle.$$

Proof. Let $\Sigma_g(\delta)$ denote the compressed surface.

$$\chi(\Sigma_g(\delta)) = (2 - 2g) + 2g = 2, \quad \Sigma_g \setminus \bigcup_j \delta_j \text{ connected} \implies \Sigma_g(\delta) \cong S^2 \implies H_\delta \cong \natural^g(S^1 \times D^2).$$

Van Kampen gives the displayed quotient, boundary epimorphism, and kernel. $\square$

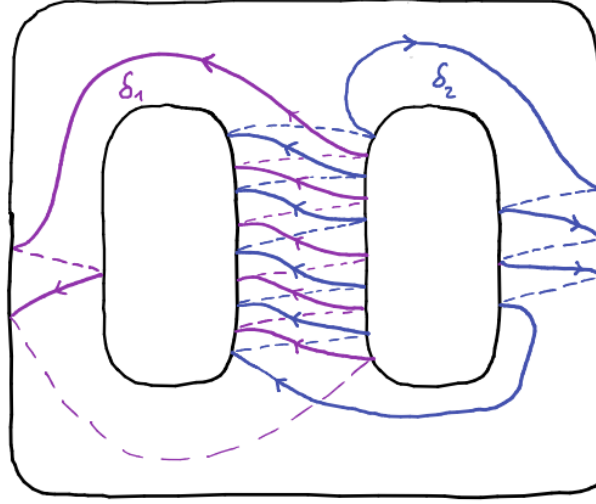


FIGURE 2. A genus-2 Heegaard diagram for the Poincaré homology sphere.

Example 1.1.3 (Poincaré homology sphere). The red and blue curves $\delta = (\delta_1, \delta_2)$ in Figure 2 form one cut system in a genus-two Heegaard diagram, up to orientation, and

$$\Sigma(2, 3, 5) \cong \text{the double cover of } S^3 \text{ branched over } P(-2, 3, 5).$$

Attaching its two 2-handles and capping the resulting sphere realizes the corresponding handlebody.

Definition 1.1.4 (Marked realization). A marked realization of $\varphi : S_g \twoheadrightarrow F_g$ is a genus- g handlebody H , an orientation-preserving $h : \Sigma_g \rightarrow \partial H$, and an isomorphism $\theta : \pi_1(H) \rightarrow F_g$ satisfying

$$\varphi = \theta \circ i_* \circ h_*, \quad i : \partial H \hookrightarrow H.$$

Two realizations are equivalent if some diffeomorphism $\Phi : H \rightarrow H'$ intertwines their boundary parametrizations up to based isotopy and satisfies $\theta' \circ \Phi_* = \theta$.

Theorem 1.1.5 (Handlebody realization, [4, 6]). *For $g \geq 1$,*

$$\varphi : S_g \twoheadrightarrow F_g \implies \varphi \text{ admits a marked realization, unique up to the equivalence of Definition 1.1.4.}$$

Proof. Let $R_g := \bigvee_{j=1}^g S^1$, choose $f : (\Sigma_g, *) \rightarrow (R_g, *)$ with $f_* = \varphi$, transverse to $q_j \in S_j^1 \setminus \{*\}$, and set $L := \bigsqcup_{j=1}^g f^{-1}(q_j)$. For the edge-dual classes η_j , surjectivity and disjointness give

$$\varphi \twoheadrightarrow f^* \text{ is injective,} \quad \text{PD}(f^* \eta_j) = [f^{-1}(q_j)], \quad C \cap C' = \emptyset \implies g \leq \text{rank} \langle [C] : C \subset L \rangle \leq g.$$

Choose independent components $\delta = (\delta_1, \dots, \delta_g) \subset L$; they form a cut system, and

$$f|_{\delta_j} \equiv q_{\ell(j)} \implies f \text{ extends over the 2-handles,} \quad \pi_2(R_g) = 0 \implies \bar{f} : H_\delta \rightarrow R_g.$$

The induced factorization satisfies

$$\psi := \bar{f}_* : \pi_1(H_\delta) \rightarrow F_g, \quad \varphi = \psi \circ (i_\delta)_*, \quad \varphi \twoheadrightarrow \implies \psi : F_g \twoheadrightarrow F_g \implies \psi \in \text{Aut}(F_g).$$

Thus (H_δ, h, ψ) is a marked realization; marked uniqueness is Jaco's splitting-homomorphism theorem. $\square$

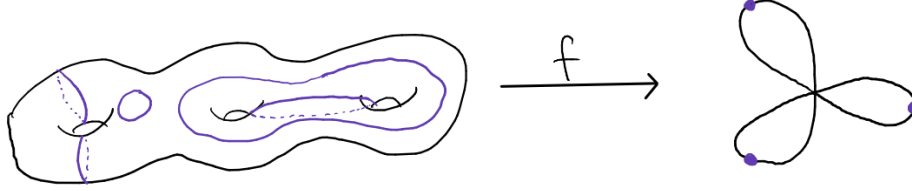


FIGURE 3. Regular-point preimages contain a realizing cut system.

Theorem 1.1.6 (Algorithmic realization, [2]). *If $\varphi : S_g \twoheadrightarrow F_g$ is specified by finite words, set*

$$L := \sum_{j=1}^g (|\varphi(x_j)| + |\varphi(y_j)|), \quad \text{a realizing cut-system diagram is constructible in time } \text{poly}(L).$$

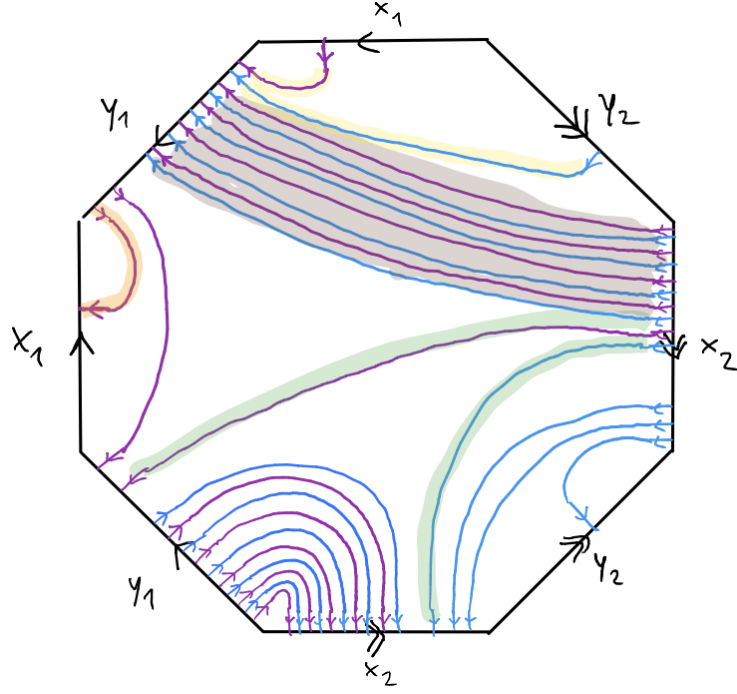
Example 1.1.7 (Genus-two reconstruction). Set $F_2 = \langle d_1, d_2 \rangle$ and $a := d_1 d_2$. Define

$$\varphi(x_1) = d_1^{-1}, \quad \varphi(y_1) = a^5 d_1^{-2}, \quad \varphi(x_2) = a^5 d_2^3, \quad \varphi(y_2) = d_2.$$

For $r := [x_1, y_1][x_2, y_2]$, the surface relation and surjectivity reduce to

$$\varphi(r) = d_1^{-1} a^5 d_1 a^{-5} \cdot a^5 d_2 a^{-5} d_2^{-1} = d_1^{-1} a^5 a a^{-5} d_2^{-1} = d_1^{-1} a d_2^{-1} = 1.$$

Moreover $d_1 = \varphi(x_1)^{-1}$ and $d_2 = \varphi(y_2)$, so $\text{im}(\varphi) = F_2$. Place the words on the octagon edges and pair inverse red $d_1^{\pm 1}$ and blue $d_2^{\pm 1}$ letters by nested arcs; the side identifications close them to a realizing cut system.

FIGURE 4. Free-cancellation pairings form the cut curves δ_1, δ_2 .

The curves recover φ by oriented intersection with the polygon edges.

1.2. Group trisections.

Definition 1.2.1 (Group trisection). Let G be finitely presented, $0 \leq k_1, k_2, k_3 \leq g$, and $k_{\alpha\beta} := k_1, k_{\beta\gamma} := k_2, k_{\gamma\alpha} := k_3$. A $(g; k_1, k_2, k_3)$ -group trisection of G is the following commutative cube of surjections, every face of which is a pushout.

$$\begin{array}{ccccc}
 \pi_1(\Sigma_g) & \xrightarrow{\quad} & F_g^\alpha & & \\
 \downarrow & \searrow & \downarrow & \searrow & \\
 & & F_g^\beta & \xrightarrow{\quad} & F_{k_{\alpha\beta}} \\
 \downarrow & & \downarrow & \downarrow & \downarrow \\
 F_g^\gamma & \xrightarrow{\quad} & F_{k_{\gamma\alpha}} & & \\
 & \searrow & \downarrow & \searrow & \\
 & & F_{k_{\beta\gamma}} & \xrightarrow{\quad} & G
 \end{array}$$

Equivalently, for cyclic $(\mu, \nu, \rho) \in \{(\alpha, \beta, \gamma), (\beta, \gamma, \alpha), (\gamma, \alpha, \beta)\}$,

$$F_g^\mu *_{S_g} F_g^\nu \cong F_{k_{\mu\nu}}, \quad F_{k_{\mu\nu}} *_{F_g^\nu} F_{k_{\nu\rho}} \cong G.$$

Definition 1.2.2 (Smooth trisection and trisection diagram). A $(g; k_1, k_2, k_3)$ -trisection of a closed connected oriented smooth four-manifold X is

$$X = X_1 \cup X_2 \cup X_3, \quad X_i \cong \natural^{k_i}(S^1 \times B^3), \quad X_i \cap X_j \cong H_g := \natural^g(S^1 \times D^2), \quad X_1 \cap X_2 \cap X_3 \cong \Sigma_g.$$

A trisection diagram is a quadruple $(\Sigma_g; \alpha, \beta, \gamma)$ of cut systems satisfying

$$H_\mu \cup_{\Sigma_g} (-H_\nu) \cong \#^{k_i}(S^1 \times S^2) \quad ((\mu, \nu, i) = (\alpha, \beta, 1), (\beta, \gamma, 2), (\gamma, \alpha, 3)), \quad \#^0(S^1 \times S^2) := S^3.$$

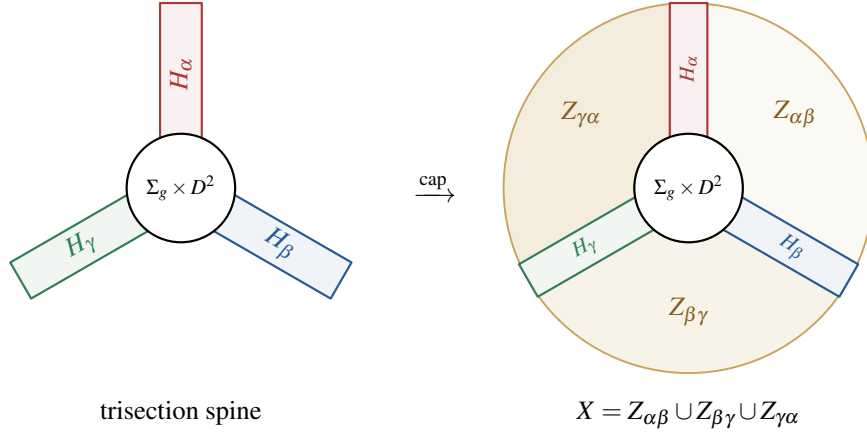


FIGURE 5. The trisection spine and its capped four-dimensional sectors.

Proposition 1.2.3 (Fundamental-group cube). *For every smooth trisection of X , its based fundamental groups form*

$$a (g; k_1, k_2, k_3)\text{-group trisection of } \pi_1(X).$$

Proof. Proposition 1.1.2 gives the boundary epimorphisms, and van Kampen gives the pushouts.

$$\begin{array}{ccccc}
 \pi_1(\Sigma_g) & \xrightarrow{\quad} & \pi_1(H_\alpha) & & \\
 \downarrow & \searrow & \downarrow & \searrow & \\
 & \pi_1(H_\beta) & \xrightarrow{\quad} & \pi_1(Z_{\alpha\beta}) & \\
 \downarrow & \downarrow & \downarrow & \downarrow & \\
 \pi_1(H_\gamma) & \xrightarrow{\quad} & \pi_1(Z_{\gamma\alpha}) & & \\
 & \searrow & \downarrow & \searrow & \\
 & \pi_1(Z_{\beta\gamma}) & \xrightarrow{\quad} & \pi_1(X) &
 \end{array}$$

Write $Z_{\alpha\beta} = X_1$, $Z_{\beta\gamma} = X_2$, and $Z_{\gamma\alpha} = X_3$. For the corresponding cyclic triples (μ, ν, i) ,

$$H_\mu = H_\mu \xrightarrow{1.1.2} \pi_1(H_\mu) \cong F_g, \quad \pi_1(H_\mu \cup_{\Sigma_g} (-H_\nu)) \cong \pi_1(H_\mu) *_{S_g} \pi_1(H_\nu) \cong F_{k_i}, \quad \pi_1(X_i) \cong F_{k_i}.$$

Applying van Kampen to the connected unions identifies the terminal colimit with $\pi_1(X)$, so every face is a pushout and the cube satisfies Definition 1.2.1. $\square$

Lemma 1.2.4 (Free three-manifold groups, [2]). *For every closed connected oriented three-manifold M ,*

$$\pi_1(M) \cong F_k \implies M \cong \#^k(S^1 \times S^2).$$

Theorem 1.2.5 (Laudenbach–Poénaru extension, [5]). *Every orientation-preserving diffeomorphism satisfies*

$$f : \#^k(S^1 \times S^2) \xrightarrow{\cong} \#^k(S^1 \times S^2) \implies f \text{ extends over } \natural^k(S^1 \times B^3), \quad \partial \natural^k(S^1 \times B^3) = \#^k(S^1 \times S^2).$$

Theorem 1.2.6 (Group-trisection correspondence, [1]). *For $k_1 = k_2 = k_3$, group trisections classify closed oriented smooth four-manifolds:*

$$\{\text{balanced group trisections}\} / (\cong, \text{stabilization}) \longleftrightarrow \{\text{closed oriented smooth 4-manifolds}\} / \text{Diff}^+.$$

2. BRIDGE TRISECTIONS AND KNOTTED-SURFACE GROUPS

Meier and Zupan place smoothly knotted surfaces in bridge position, while Blackwell–Kirby–Klug–Longo–Ruppik encode the resulting tangles by bounding homomorphisms and pushout cubes [7, 8, 2]. All fundamental groups below use a basepoint on the central punctured surface and fixed whiskers to the curves, arcs, and punctures.

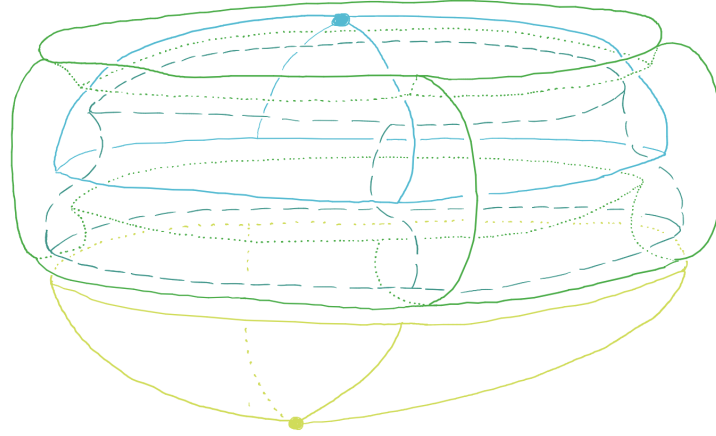


FIGURE 6. A movie presentation of the spun trefoil.

2.1. Bridge position and shadows. Figure 6 displays the spun trefoil through three trivial tangle sectors, despite the knottedness of the closed surface.

Definition 2.1.1 (Trivial tangle and shadow). Let H be a genus- g handlebody with $\partial H = \Sigma_g$. A b -strand trivial tangle and its shadow are

$$T = T_1 \sqcup \cdots \sqcup T_b \subset H, \quad B_i \cap B_j = \emptyset \ (i \neq j), \quad \partial B_j = T_j \cup S_j,$$

where the T_j are properly embedded arcs, the B_j are bridge disks, and $S = \{S_1, \dots, S_b\}$ with $S_j \subset \Sigma_g$ and $\partial S_j = \partial T_j$. A curve-and-arc diagram (C, S) has a cut system $C = \{C_1, \dots, C_g\}$ for H with $C \cap S = \emptyset$. A bubble disk in $H \setminus T$ encloses one B_j and separates T_j from $T \setminus T_j$.

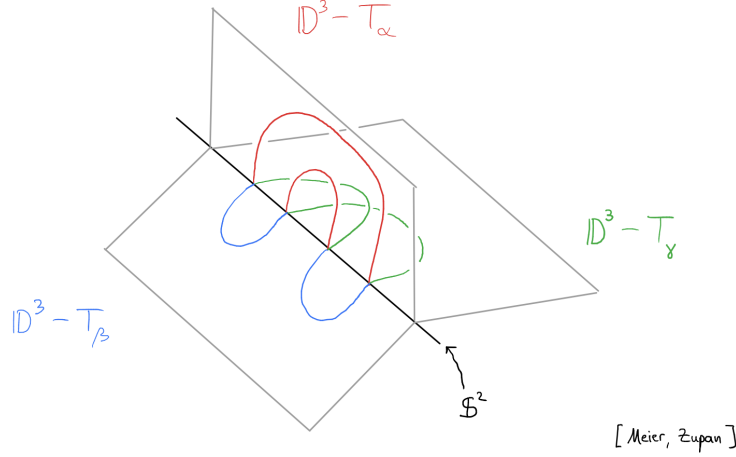


FIGURE 7. Three trivial tangles and their shadows encode a bridge trisection.

Proposition 2.1.2 (Tangle-complement group). *For a trivial b -strand tangle $T \subset H$, order and orient C and S . Boundary inclusion and disk-dual generators give*

$$H \setminus T \simeq \bigvee^{g+b} S^1, \quad \pi_1(H \setminus T, *) \cong F_{g+b} = \langle t_1, \dots, t_b, h_1, \dots, h_g \rangle, \quad \iota_*: \pi_1(\Sigma_g \setminus \partial T, *) \twoheadrightarrow \pi_1(H \setminus T, *).$$

For an oriented transverse loop λ , label each crossing by $z_r \in \{t_1, \dots, t_b, h_1, \dots, h_g\}$ and give a crossing with $A \subset C \cup S$ sign $+1$ exactly when $(\dot{\lambda}, \dot{A})$ orients Σ_g . Then

$$\lambda \cap (C \cup S) = (z_1^{\epsilon_1}, \dots, z_m^{\epsilon_m}), \quad \epsilon_r \in \{-1, 1\}, \quad \iota_*([\lambda]) = \prod_{r=1}^m z_r^{\epsilon_r}.$$

Proof. Cutting along the cut and bridge disks gives a ball, hence $H \setminus T \simeq \bigvee^{g+b} S^1$. Its dual loops satisfy $h_i \cdot C_k = \delta_{ik}$, $t_j \cdot S_\ell = \delta_{j\ell}$, and all cross-intersections vanish. The dual loops lie in the boundary image, while signed crossings read the displayed word; hence ι_* is onto and the formula follows. $\square$

Definition 2.1.3 (Bridge trisection, [7, 8]). A $(g; k_\alpha, k_\beta, k_\gamma; b; c_\alpha, c_\beta, c_\gamma)$ -bridge trisection of $K \subset X^4$ is

$$(X, K) = (X_\alpha, \mathcal{D}_\alpha) \cup (X_\beta, \mathcal{D}_\beta) \cup (X_\gamma, \mathcal{D}_\gamma), \quad X_\mu \cong \natural^{k_\mu}(S^1 \times B^3).$$

For cyclic (μ, ν, ρ) , set $H_\mu := X_\nu \cap X_\rho$ and $T_\mu := K \cap H_\mu$. Then

$$\mathcal{D}_\mu = K \cap X_\mu \text{ is } c_\mu \text{ boundary-parallel disks,} \quad H_\mu \cong H_g, \quad T_\mu \text{ is a trivial } b\text{-tangle.}$$

$$\Sigma_g = X_\alpha \cap X_\beta \cap X_\gamma, \quad P = K \cap \Sigma_g, \quad |P| = 2b, \quad \chi(K) = c_\alpha + c_\beta + c_\gamma - b.$$

For the standard genus-zero trisection of S^4 , write $(b; c_\alpha, c_\beta, c_\gamma)$; here $X_\mu \cong B^4$, $H_\mu \cong D^3$, and $\Sigma_g = S^2$.

By Figure 7, the bridge trisection is determined by its three-tangle spine together with the ambient trisection.

Theorem 2.1.4 (Bridge position and stable equivalence, [7, 8, 3]). *For a smoothly embedded closed surface K and bridge trisections $\mathcal{T}_0, \mathcal{T}_1$ of isotopic surfaces,*

$$K \simeq K' \text{ in bridge position,} \quad \mathcal{T}_0 \sim \mathcal{T}_1 \text{ after ambient stabilization and perturbation/deperturbation.}$$

2.2. Bounding homomorphisms and reconstruction. Let $P = \{p_1, \dots, p_{2b}\} \subset \Sigma_g$, where the same symbols denote negatively oriented meridians based by the chosen whiskers. With $[a, b] = aba^{-1}b^{-1}$, set

$$\Pi_{g,b} := \pi_1(\Sigma_g \setminus P, *) = \langle p_1, \dots, p_{2b}, a_1, b_1, \dots, a_g, b_g \mid p_1 \cdots p_{2b} = [a_1, b_1] \cdots [a_g, b_g] \rangle.$$

Definition 2.2.1 (Bounding homomorphism, [2]). Let $\phi: \Pi_{g,b} \twoheadrightarrow F_{g+b} = \langle t_1, \dots, t_b, h_1, \dots, h_g \rangle$ be an epimorphism. It is bounding when, for the quotient q killing $t_1, \dots, t_b$,

$$q: F_{g+b} \twoheadrightarrow \langle h_1, \dots, h_g \rangle, \quad \langle q\phi(a_1), q\phi(b_1), \dots, q\phi(a_g), q\phi(b_g) \rangle = \langle h_1, \dots, h_g \rangle,$$

and there are a bijection and conjugating words

$$\varepsilon: \{p_1, \dots, p_{2b}\} \longrightarrow \{t_1, t_1^{-1}, \dots, t_b, t_b^{-1}\}, \quad u_i \in F_{g+b}, \quad \phi(p_i) = u_i \varepsilon(p_i) u_i^{-1} \quad (1 \leq i \leq 2b).$$

Theorem 2.2.2 (Bounding realization, [2]). *If a bounding homomorphism ϕ has total word length L , then*

$$\phi \text{ has a unique rel-}\partial H \text{ realization } (H, T), \quad (C, S) \text{ is constructible in time poly}(L).$$

Example 2.2.3 (A shadow map). The oriented shadow in Figure 8 induces $\theta: \Pi_{0,2} = \langle a_1, a_2, a_3, a_4 \mid a_1 a_2 a_3 a_4 = 1 \rangle \twoheadrightarrow F_2 = \langle x, y \rangle$. Write $v_i := \theta(a_i)$. Then

$$(v_1, v_2, v_3, v_4) := (x^{-1}, yx^{-1}y^{-1}xy^{-1}, yx^{-1}xy^{-1}xy^{-1}, y), \quad \theta \text{ is bounding.}$$

Proof. The shadow gives the displayed words, and free reduction gives

$$v_1 v_2 v_3 v_4 = x^{-1} y x^{-1} y^{-1} x x^{-1} y x y^{-1} x = x^{-1} y x^{-1} x y^{-1} x = x^{-1} y y^{-1} x = 1.$$

Set $u := yx^{-1}$. Since $v_1 = x^{-1}$ and $v_4 = y$, the map is onto, while $v_2 = uy^{-1}u^{-1}$ and $v_3 = (uy)x(uy)^{-1}$. Thus the central letters are x^{-1}, y^{-1}, x, y , so Definition 2.2.1 proves the claim. $\square$

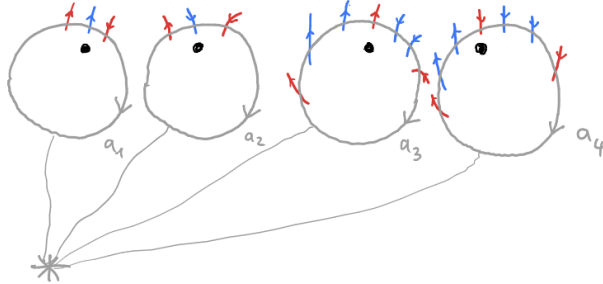


FIGURE 8. Signed shadow intersections give $v_1, \dots, v_4$.

Example 2.2.4 (Four-punctured-sphere reconstruction). Let $\Pi_{0,2} = \langle a_1, a_2, a_3, a_4 \mid a_1 a_2 a_3 a_4 = 1 \rangle$ and $F_2 = \langle x, y \rangle$. Define $\phi: \Pi_{0,2} \twoheadrightarrow F_2$, and write $w_i := \phi(a_i)$. Then

$$(w_1, w_2, w_3, w_4) := (yxy^{-1}, yx^{-1}y^{-1}, yx^2yx^{-2}y^{-1}, yx^2y^{-1}x^{-2}y^{-1}), \quad \phi \text{ is bounding.}$$

Proof. Since $w_2 = w_1^{-1}$ and $w_4 = w_3^{-1}$, the surface relation holds. Moreover,

$$w_3 = w_1^2 y w_1^{-2} \implies y = w_1^{-2} w_3 w_1^2, \quad x = y^{-1} w_1 y \implies \langle w_1, w_2, w_3, w_4 \rangle = F_2.$$

The four formulas exhibit conjugates with central letters x, x^{-1}, y, y^{-1} , so Definition 2.2.1 gives the bounding tangle. $\square$

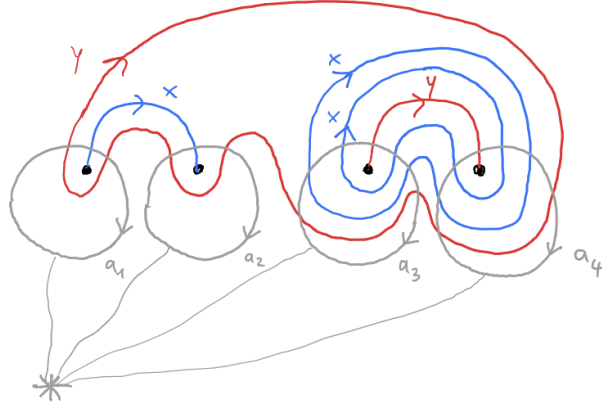


FIGURE 9. Signed shadow intersections give the epimorphism to $\langle x, y \rangle$.

Figure 9 records the signed crossings and the completed reconstruction for Example 2.2.4.

Definition 2.2.5 (Stallings fold, [10]). For a directed basis-labeled graph $\Gamma \rightarrow R_n$, a fold is

$$e_1, e_2 \text{ equally labeled, } \partial_- e_1 = \partial_- e_2 \text{ or } \partial_+ e_1 = \partial_+ e_2 \implies e_1 \sim e_2,$$

with the other endpoints identified when distinct.

Fact 2.2.6 (Stallings criterion, [10]). For $v_1, \dots, v_m \in F_n$, let $\Gamma(v_1, \dots, v_m)$ be the based labeled graph obtained by reading each word around one wedge petal. Then

$$\langle v_1, \dots, v_m \rangle = F_n \iff \Gamma(v_1, \dots, v_m) \longrightarrow \dots \longrightarrow R_n \text{ by folds of the core graph.}$$

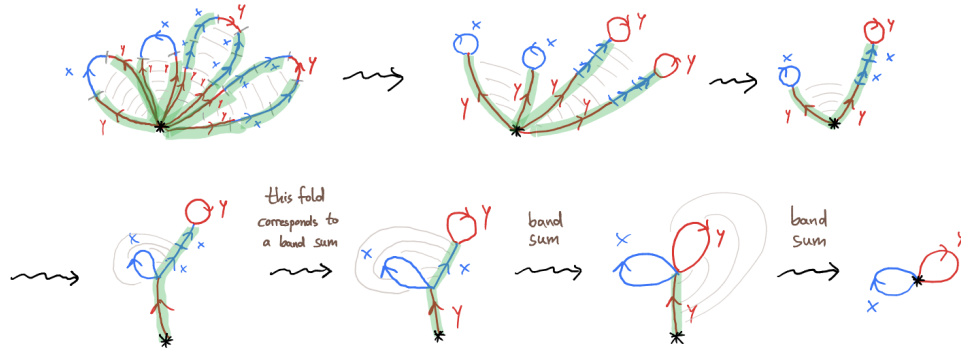


FIGURE 10. Stallings folds correspond to shadow band sums.

By Figure 10, the folds and band sums give

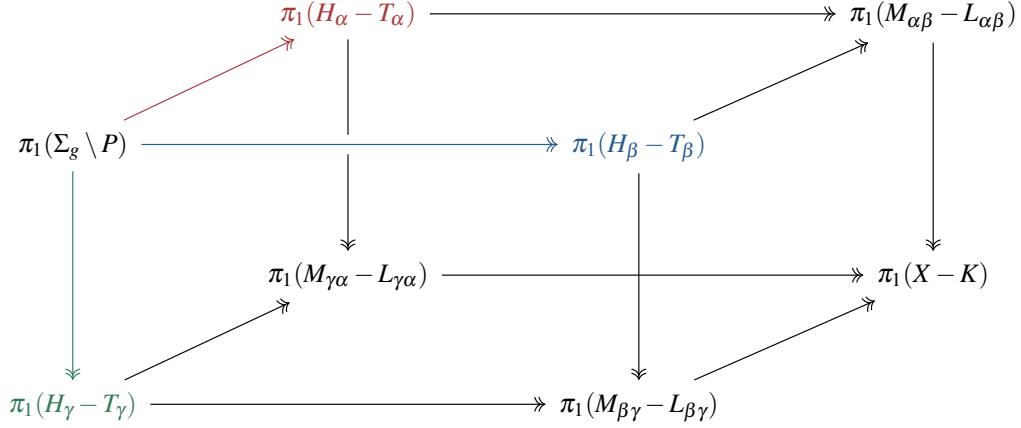
$$\Gamma(w_1, w_2, w_3, w_4) \longrightarrow \Gamma_1 \longrightarrow \dots \longrightarrow R_2, \quad \langle w_1, w_2, w_3, w_4 \rangle = \langle x, y \rangle.$$

2.3. The complement cube. Let (X, K) have a based, parametrized $(g; k_\alpha, k_\beta, k_\gamma; b; c_\alpha, c_\beta, c_\gamma)$ -bridge trisection with $b \geq 1$. For distinct $\mu, \nu \in \{\alpha, \beta, \gamma\}$, set $M_{\mu\nu} := H_\mu \cup_{\Sigma_g} (-H_\nu)$, $L_{\mu\nu} := T_\mu \cup_P T_\nu \subset M_{\mu\nu}$, $G_{\mu\nu} := \pi_1(M_{\mu\nu} \setminus L_{\mu\nu}, *)$, and $G_K := \pi_1(X \setminus K, *)$. Then

$$G_\Sigma := \pi_1(\Sigma_g \setminus P, *) \cong F_{2g+2b-1}, \quad G_\mu := \pi_1(H_\mu \setminus T_\mu, *) \cong F_{g+b}.$$

Use $k_{\alpha\beta} := k_\gamma$, $k_{\beta\gamma} := k_\alpha$, $k_{\gamma\alpha} := k_\beta$ and $c_{\alpha\beta} := c_\gamma$, $c_{\beta\gamma} := c_\alpha$, $c_{\gamma\alpha} := c_\beta$.

Definition 2.3.1 (Knotted-surface group cube). These maps form the knotted-surface group cube.



Every arrow is surjective and every square face is a pushout. Moreover

$$M_{\mu\nu} \cong \#^{k_{\mu\nu}}(S^1 \times S^2), \quad L_{\mu\nu} \text{ is a } c_{\mu\nu}\text{-component unlink}, \quad G_{\mu\nu} \cong F_{k_{\mu\nu} + c_{\mu\nu}}.$$

For the standard trisection of S^4 , its four layers are

$$F_{2b-1} \longrightarrow (F_b)^3 \longrightarrow (F_{c_{\alpha\beta}}, F_{c_{\beta\gamma}}, F_{c_{\gamma\alpha}}) \longrightarrow G_K = \pi_1(S^4 \setminus K, *).$$

Fact 2.3.2 (Free group detects unlinks, [2]). *For an n -component link $L \subset \#^k(S^1 \times S^2)$,*

$$\pi_1(\#^k(S^1 \times S^2) \setminus L, *) \cong F_{n+k} \implies L \text{ is the } n\text{-component unlink}.$$

The case $k = 0$ is the S^3 unlink criterion.

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